SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	Vinil Kumar S	**Reg. No.:**	192511226
Section / Batch:		**Date:**	20/07/2026

Code of Conduct

I Vinil Kumar S (192511226) *(Name / Reg No)*
certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S. Vinil Kumar

Instructions

- This test contains 5 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
	GRAND TOTAL:		**100 Marks**

38	192524251	VAISHNAV M	1. Two algorithms solve the same problem. Algorithm A uses recursion, while Algorithm B uses an explicit stack. Compare both approaches in terms of memory usage, execution flow, and risk of stack overflow. 2. A stack has a capacity of 5 elements. Explain what happens if the following operations are executed: Push(A), Push(B), Push(C), Push(D), Push(E), Push(F). Identify the error and suggest two possible solutions.
39	192525187	VEERA SANTHOSH A	1. Compare the performance of a linear queue and a circular queue when 100 enqueue and dequeue operations are performed alternately. 2. A queue implemented using an array report "Queue Full" even though some positions are empty. ➢ Analyze why this occurs. ➢ Explain how a circular queue solves this problem.
40	192511226	VINIL KUMAR S	1. Consider the singly linked list: 10 → 20 → 30 → 40 → 50 Delete the node containing 30. Explain how the links change after deletion. 2. Given two sorted linked lists: L1: 1 → 3 → 5 → 7 L2: 2 → 4 → 6 → 8

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature:	Signature: ____________	Signature: ____________
Date: ____________	Date: ____________	Date: ____________

1. Consider the singly linked list : 10 → 20 → 30 → 40 → 50
Delete the node containing 30. Explain how the links change after deletion.

Ans:

A singly linked list contains the elements :

10 → 20 → 30 → 40 → 50.

The task is to delete the node containing 30 and update the links so that the list remains connected. Since it is a singly linked list, each node contains data and a pointer to the next node.

Constraints :-

→ The node to be deleted exists in the list

→ The list should remain Connected after deletion.

Before deletion :-

Head

[10] → [20] → [30] → [40] → [50]

↑

Delete this node.

After changing the link :-

Head

[10] → [20] ——————— → [40] → [50]

(30 removed)

Final linked list :-

[10] → [20] → [40] → [50] → Null
Head.

The next pointer of node 20 is changed to point directly to 40, bypassing node 30. Then node 30 is deleted.

Given two sorted linked lists :

L1 : 1 → 3 → 5 → 7

L2 : 2 → 4 → 6 → 8.

Ans:-

Given : L1: 1 → 3 → 5 → 7
L2: 2 → 4 → 6 → 8

* To merge the two sorted linked lists into a single sorted linked list.

⇒ Compare 1 and 2 → Choose 1

Merge list :
1

⇒ Compare 3 and 2 → Choose 2
Merged list :
1 → 2

⇒ Compare 3 and 4 → Choose 3
Merged list :
1 → 2 → 3

Compare 5 and 4 → choose 4

Merged list:

1 → 2 → 3 → 4

⇒ Compare 5 and 6 → choose 5

Merged list:

1 → 2 → 3 → 4 → 5.

⇒ Compare 7 and 6 → choose 6

Merged list:

1 → 2 → 3 → 4 → 5 → 6.

⇒ Compare 7 and 8 → choose 7

Merged list:

1 → 2 → 3 → 4 → 5 → 6 → 7

⇒ L1 is empty, attach remaining node 8.

Final list:

1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → Null.

Algorithm:-

1. Compare the first nodes of both lists
2. Insert the smaller node into the merged list
3. Move the pointer of the selected list
4. Repeat until one list becomes empty.
5. Attach the remaining nodes of the other list.

Final list :

after merging two sorted linked list into single sorted linked list ,

1 → 2 → 3 → 4 → 5 → 6 → 7 → 8 → Null